Grade 10 – Term 1: C6 & C7

Expected Monetary Value and Sensitivity Analysis

1. Museum Gallery. A museum must choose an expanded gallery (A_1) or a compact gallery (A_2) before regional tourism is known. Tourism may be busy (S_1) or quiet (S_2). Visitor records give estimated probabilities of 70% and 30%, respectively. All payoffs and EMVs are in thousand USD.

Alternative	S_1	S_2
$P(S_j)$:	70%	30%
Expanded gallery (A_1)	62	8
Compact gallery (A_2)	39	26

Calculate the EMV of each alternative and select an alternative using the Bayes decision criterion. Show your work.

2. Orchard Lease. A grower must lease a hillside orchard (A_1) or a valley orchard (A_2) before seasonal rainfall is known. Rainfall may be ample (S_1), moderate (S_2), or scarce (S_3). Weather records give estimated probabilities of 40%, 35%, and 25%, respectively. Payoffs and EMVs are in thousand USD.

	Rainfall		
Alternative	Ample (S_1)	Moderate (S_2)	Scarce (S_3)
Hillside (A_1)	96	34	-18
Valley (A_2)	68	48	24

Calculate the EMV of each alternative and select an alternative using the Bayes decision criterion. Show your work.

3. Campsite Plan. A campsite owner must choose open pitches (A_1), covered pitches (A_2), cabins (A_3), or a fixed lease (A_4) before the summer weather is known. Weather records estimate dry weather (S_1) at 55% and wet weather (S_2) at 45%. The fixed lease guarantees the same payoff in either state. Payoffs and EMVs are in thousand USD.

	Weather	
Alternative	Dry (S_1)	Wet (S_2)
Open pitches (A_1)	94	-16
Covered pitches (A_2)	70	18
Cabins (A_3)	46	32
Fixed lease (A_4)	24	24

Calculate the EMV of each alternative and select an alternative using the Bayes decision criterion. Show your work.

4. Translation Rights. A publisher must choose worldwide translation rights (A_1), selected language rights (A_2), regional rights (A_3), or a fixed license (A_4) before overseas interest in a novel is known. Interest may be strong (S_1), steady (S_2), or weak (S_3). Reader research estimates probabilities of 45%, 25%, and 30%, respectively. The fixed license guarantees the same payment in all states. Payoffs and EMVs are in thousand USD.

	Overseas interest		
Alternative	S_1	S_2	S_3
A_1	78	42	12
A_2	62	49	23
A_3	44	37	31
A_4	26	26	26

Calculate the EMV of each alternative and select an alternative using the Bayes decision criterion. Show your work.

5. Irrigation Plan. A greenhouse cooperative must select intensive irrigation (A_1), balanced irrigation (A_2), or minimal irrigation (A_3) before the dry season. The external water supply may be abundant (S_1), adequate (S_2), limited (S_3), or restricted (S_4). Reservoir research estimates the probabilities as 25%, 30%, 25%, and 20%, respectively. Payoffs and EMVs are in thousand USD.

	Water supply			
Alternative	S_1	S_2	S_3	S_4
A_1	102	76	34	-14
A_2	79	63	41	21
A_3	52	47	39	29

Calculate the EMV of each alternative and select an alternative using the Bayes decision criterion. Show your work.

6. Wildlife Hide. A reserve must choose an open hide (A_1) or a sheltered hide (A_2) before the season's visibility is known. Visibility may be clear (S_1) or misty (S_2). Let p be the reserve manager's subjective probability of clear visibility, based on judgment; mist has probability $1 - p$, where $0 \leq p \leq 1$. Payoffs and EMVs are in thousand USD.

Alternative	S_1	S_2
$P(S_j)$:	p	$1 - p$
Open hide (A_1)	68	8
Sheltered hide (A_2)	48	38

Construct the EMV function for each alternative. Use the Bayes decision criterion to determine the preferred alternative throughout the feasible probability domain, identifying all optimal ties and decision boundaries. Show your work.

7. Art Workshop. An art school must book a courtyard (A_1) or an indoor studio (A_2) before workshop-day weather is known. Weather may be dry (S_1) or rainy (S_2). Let p be the coordinator's subjective probability of dry weather, based on judgment; rain has probability $1 - p$, with $0 \leq p \leq 1$. Payoffs and EMVs are in thousand USD.

Alternative	S_1	S_2
$P(S_j)$:	p	$1 - p$
Courtyard (A_1)	58	6
Indoor studio (A_2)	43	31

Construct the EMV function for each alternative. Use the Bayes decision criterion to determine the preferred alternative throughout the feasible probability domain, identifying all optimal ties and decision boundaries. Show your work.

8. Apiary. A beekeeper must choose a meadow apiary (A_1) or woodland apiary (A_2) before flowering conditions are known: favorable (S_1) or poor (S_2). Let p be the beekeeper's subjective probability of favorable conditions, based on judgment; poor conditions have probability $1 - p$, $0 \leq p \leq 1$. Payoffs and EMVs are in thousand USD.

Alternative	S_1	S_2
$P(S_j)$:	p	$1 - p$
Meadow apiary (A_1)	76	12
Woodland apiary (A_2)	51	32

Construct the EMV function for each alternative. Use the Bayes decision criterion to determine the preferred alternative throughout the feasible probability domain, identifying all optimal ties and decision boundaries. Show your work.

9. Archive. An archive must choose full service (A_1) or basic service (A_2) before grant funding is known: ample (S_1) or limited (S_2). Let p be the director's subjective probability of ample funding, based on judgment; limited funding has probability $1 - p$, $0 \leq p \leq 1$. Payoffs and EMVs are in thousand USD.

Alternative	S_1	S_2
$P(S_j)$:	p	$1 - p$
Full service (A_1)	57	44
Basic service (A_2)	49	36

Construct the EMV function for each alternative. Use the Bayes decision criterion to determine the preferred alternative throughout the feasible probability domain, identifying all optimal ties and decision boundaries. Show your work. Does a feasible switching threshold exist? Justify your conclusion.

10. Vineyard Layout. A vineyard must choose hillside vines (A_1), mixed plots (A_2), or sheltered vines (A_3) before growing conditions are known. Conditions may be favorable (S_1) or adverse (S_2). Let p be the grower's subjective probability of favorable conditions, based on judgment; adverse conditions have probability $1 - p$, where $0 \leq p \leq 1$. Payoffs and EMVs are in thousand USD.

Alternative	S_1	S_2
$P(S_j)$:	p	$1 - p$
Hillside vines (A_1)	86	6
Mixed plots (A_2)	66	26
Sheltered vines (A_3)	46	36

Construct the EMV function for each alternative. Use the Bayes decision criterion to determine the preferred alternative throughout the feasible probability domain, identifying all optimal ties and decision boundaries. Show your work. Determine whether each pairwise intersection changes the optimal decision.

11. Archaeological Survey. A survey firm must choose extensive fieldwork (A_1) or a limited survey (A_2) before land-access permits are decided. Access may be full (S_1), partial (S_2), or denied (S_3). Let p be the director's subjective probability of full access, based on judgment, with $0 \le p \le 1$. Conditional on access not being full, the director assigns probability $\alpha = 0.25$ to partial access and $1 - \alpha = 0.75$ to denial. Thus the state probabilities are p, $0.25(1-p)$, and $0.75(1-p)$. Payoffs and EMVs are in thousand USD.

Alternative	S_1	S_2	S_3
$P(S_j)$:	p	$0.25(1-p)$	$0.75(1-p)$
Extensive (A_1)	92	60	12
Limited (A_2)	62	44	28

Construct the EMV function for each alternative. Use the Bayes decision criterion to determine the preferred alternative throughout the feasible probability domain, identifying all optimal ties and decision boundaries. Show your work.

12. Mineral Exploration. An exploration firm must choose deep drilling (A_1), mixed drilling (A_2), shallow drilling (A_3), or a surface survey (A_4) before the site's mineral grade is known. Grade may be rich (S_1) or lean (S_2). Let p be the geologist's subjective probability of rich grade, based on judgment; lean grade has probability $1 - p$, with $0 \le p \le 1$. Payoffs and EMVs are in thousand USD.

Alternative	S_1	S_2
$P(S_j)$:	p	$1-p$
Deep drilling (A_1)	78	8
Mixed drilling (A_2)	73	23
Shallow drilling (A_3)	63	33
Surface survey (A_4)	48	38

Construct the EMV function for each alternative. Use the Bayes decision criterion to determine the preferred alternative throughout the feasible probability domain, identifying all optimal ties and decision boundaries. Show your work. Determine whether each pairwise intersection changes the optimal decision.

13. Marina Berth Agreement. A marina must choose a seasonal berth agreement (A_1) or an annual berth agreement (A_2) before coastal sailing conditions are known. Conditions may be excellent (S_1), fair (S_2), or stormy (S_3). Let p and q be the manager's subjective probabilities of excellent and fair conditions, based on judgment. Stormy conditions have probability $1 - p - q$. The feasible domain is

$$p \geq 0, \qquad q \geq 0, \qquad p + q \leq 1.$$

Payoffs and EMVs are in thousand USD.

Alternative	S_1	S_2	S_3
$P(S_j)$:	p	q	$1 - p - q$
Seasonal berths (A_1)	84	56	9
Annual berths (A_2)	54	41	29

Construct the EMV function for each alternative. Use the Bayes decision criterion to determine the preferred alternative throughout the feasible probability domain, identifying all optimal ties and decision boundaries. Show your work. Identify the indifference line and the region where each agreement is preferred.